# Finite terms ($\Lambda \to 0$, $\vee \to \infty$) for a Gaussian projectile: three-$\rho$ rows reduced to two-$\rho$ parts, everything in colour-singlet form

Input: your `Limiting_cases.tex`, in the simplified form of `Limiting_terms.pdf` (LT). Scripts: `singlet3.py` (reduction, projection, checks), `srender3.py` (the traces below, generated, not typed).

## 1 Answer

1. **Gaussian projectile.** $\langle\rho\rangle = 0$, $\langle\rho^a(u)\rho^b(v)\rangle = \delta^{ab}\mathcal{P}(u,v)$, and the symmetrized three-point function vanishes. Each three-$\rho$ row is $S(\mathcal{O}) + \mathcal{O}^{(2)}$; $\langle S(\mathcal{O})\rangle = 0$, so only $\mathcal{O}^{(2)}$ is left: one commutator, i.e. two charges merged into one point (Sec. 2).

2. **Colour singlet.** With $\mathcal{P}(u,v) = \frac{\langle\rho^f(u)\rho^f(v)\rangle}{N_c^2-1}$ every colour factor is a closed network of adjoint Wilson lines, written with traces: $\mathrm{Tr}\{UU^\dagger\}$, $\mathrm{Tr}\{T^eAT^eB\}$ ($(T^e)_{bc} = -if^{ebc}$).

3. **Result:** Eq. (6.1). The finite terms are bilinear in $\mathcal{P}$; their colour is

   - dipole-like traces $\mathrm{Tr}\{U(p)U^\dagger(q)\}$, with coefficient $N_c$;
   - quadrupole-like $\mathrm{Tr}\{T^eAT^eB\}$ (two adjoint dipoles joined by a gluon).

   The kernels are those of LT, with the merged point set equal.

4. **Checked** (general SU(3) adjoint $U$, not symmetric):

   - the traces equal the direct colour contraction ($\leq 8\times10^{-16}$);
   - the pair rule equals the reordering engine of the earlier notes ($\leq 2.4\times10^{-16}$);
   - the $y' \to z$ colour operators of Group III equal the original ones at $U(y') = U(z)$ (exact).

## 2 Rules

**Reduction.** Write a three-$\rho$ colour operator with your charge order, $\mathcal{O} = T^{a_1a_2a_3}\rho^{a_1}(u_1)\rho^{a_2}(u_2)\rho^{a_3}(u_3)$ ($\{B,C\}$ written out). Up to terms with one $\rho$, which vanish for $\langle\rho\rangle = 0$,

$$\mathcal{O} = S(\mathcal{O}) + \mathcal{O}^{(2)}, \qquad \mathcal{O}^{(2)} = \frac{1}{2}\sum_{i<j}\mathcal{O}\Big|_{\rho^{a_i}(u_i)\rho^{a_j}(u_j)\to[\rho^{a_i}(u_i),\rho^{a_j}(u_j)]}, \qquad [\rho^a(u),\rho^b(v)] = if^{abc}\rho^c(u)\,\delta^{(2)}(u-v). \tag{2.1}$$

$(ABC - \sigma(ABC))$ is the sum of the commutators of the pairs that $\sigma$ inverts; averaged over the six orders each pair is inverted half of the time.) $S$ is the fully symmetric part. For a colour singlet $\langle S(\rho\rho\rho)\rangle \propto d^{abc}\langle d\rho\rho\rho\rangle$, $f^{abc}\langle fS(\rho\rho\rho)\rangle$; both vanish for a Gaussian.

**Kernels.** The $\delta^{(2)}(u_i - u_j)$ removes the integral over $u_j$. The kernel of the row is evaluated at $u_j = u_i$: written $\mathcal{K}\big|_{u_j\to u_i}$. The third charge $u_k$ stays. So

$$\left\langle\frac{d^3N_{\rm row}}{d^3k}\right\rangle = \frac{1}{(2\pi)^3}\int\sum_{\rm merges}\mathcal{K}_{\rm row}\big|_{u_j\to u_i}\,\mathcal{P}(u_i,u_k)\,C^{\rm row}_{u_j\to u_i}, \tag{2.2}$$

with $C$ the projected colour factor of that merge (merges not listed are zero, e.g. the pair inside an anticommutator).

**Index placement.** As in `Finite_terms_singlet.pdf`, Sec. 2.2. Group I as written; Groups II and III with the general-$U$ placement of correction M4 (identical to yours for $U = U^T$).

**c.c.** After the projection every trace is real ($U$ real; two $T^e$'s give a real product) and $\mathcal{P}$ is real and symmetric. So for Groups I and III

$$\text{row} + \text{c.c.} = 2\,\mathrm{Re}\,(\text{row}). \tag{2.3}$$

In the $\mathcal{K}_{\rm IV}$ family the Group II Row I part 1 term enters with its block A; block B ($1b$) is its complex conjugate, so 2Re covers it too.

# 3 Kernels (from LT, $d_\perp = 2$)

All kernels include their prefactor, phase and longitudinal integral; the transverse integrals are those of the row.

$$\mathcal{K}_{\rm II} = -\frac{g^4}{64\pi^6 k^+}\, e^{-ik\cdot(w'-w)} \int d^2p\, d^2q\; e^{iq\cdot(w-y)} e^{ip\cdot(y-x)}\, \frac{(x'-w')^i\, R^i(\mathbf{q},\mathbf{p})}{(x'-w')^2\,\mathbf{p}^2\mathbf{q}^2},$$

$$R^i = -2q^i \log\frac{p^2}{q^2} + \frac{(2p^2-2q\cdot p+q^2)q^i + q^2 p^i}{(q-p)^2}\log\frac{q^2}{(q-p)^2} \tag{3.1}$$

$$\mathcal{K}_{\rm III} = \text{the sum of the four parts of your Row III (LT: unchanged) without the colour bracket,} \tag{3.2}$$

$$\mathcal{K}_{\rm IV} = \frac{ig^4}{8\pi^5 k^+} e^{-ik\cdot(w'-w)} \frac{(x'-w')^i (y-z)^j}{(x'-w')^2(y-z)^2} \log\frac{(x-z)^2}{(x-w)^2}\Big[\frac{(x-z)^j(z-w)^i}{(z-w)^2(x-z)^2} - \frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2} - \frac{(x-z)^j(x-w)^i}{(x-z)^2(x-w)^2} + \frac{\delta^{ij}}{2(z-w)^2}\Big], \tag{3.3}$$

$$\mathcal{K}_{\rm V1} = \frac{ig^4}{32\pi^5 k^+}\int_0^1 d\xi\, e^{-ik\cdot(w'-(1-\xi)z-\xi x)} \frac{(x'-w')^i (z-x)^m (y-z)^k (y'-x)^j}{(x'-w')^2(z-x)^2(y-z)^2(y'-x)^2}\Big[\delta_{im}\delta_{jk} - \Big[\frac{1}{\xi}\Big]_+\delta_{jm}\delta_{ik} - \Big[\frac{1}{1-\xi}\Big]_+\delta_{ij}\delta_{mk}\Big], \tag{3.4}$$

$$\mathcal{K}_C = \frac{ig^4}{16\pi^5 k^+}\int_0^1 d\zeta\, e^{-i\frac{k}{\zeta}\cdot(y'-\zeta w-(1-\zeta)z)} \times \frac{\frac{1}{\zeta}(y'-z)\cdot(x'-y')(y-w)\cdot(x-z) - \big(\big[\frac{1}{1-\zeta}\big]_+ + \frac{1}{\zeta}\big)(y'-z)\cdot(x-z)(x'-y')\cdot(y-w)}{(y'-z)^2(x'-y')^2(y-w)^2(x-z)^2}, \tag{3.5}$$

$$\mathcal{K}_{C2} = \frac{ig^4}{16\pi^5 k^+}\int_0^1 \frac{d\zeta}{\zeta^2}\, e^{-i\frac{k}{\zeta}\cdot(y'-\zeta w-(1-\zeta)z)}\, \frac{(y'-z)\cdot(y-w)\,(x-z)\cdot(x'-y')}{(y'-z)^2(y-w)^2(x-z)^2(x'-y')^2}, \qquad \mathcal{K}_{C2z} = \mathcal{K}_{C2}\Big|_{\frac{(x'-y')}{(x'-y')^2}\to\frac{(x'-z)}{(x'-z)^2}}. \tag{3.6}$$

(In $\mathcal{K}_{\rm II}$ the second momentum is written $q$: your $\int d^2k$ uses the letter of the measured gluon.) Charges: Group I Rows II, III, IV and Group III: $(x', x, y)$; Group I Row V part 1: $(x', y, y')$.

- $\mathcal{K}_{\rm IV}$ **family:** Group I Row IV + Group III Rows I, II + Group II Row I part 1 $= \frac{1}{(2\pi)^3}\int \mathcal{K}_{\rm IV}\big(\mathcal{O}_{\rm IV} - \mathcal{O}_{\rm G3I} - \mathcal{O}_{\rm G3II} - \frac{1}{2}\mathcal{O}_A\big)$ (+c.c.).
- $\mathcal{K}_C$ **family:** Group III Rows III part 1, V, VI $= \frac{1}{(2\pi)^3}\int \big[\mathcal{K}_C\mathcal{O}_C - \mathcal{K}_{C2}\mathcal{O}_C + \mathcal{K}_{C2z}\mathcal{O}_C^z\big]$ (+c.c.), where:
  - $\mathcal{O}_C = \mathcal{O}_{\rm III(1)} + 2\mathcal{O}_{\rm V} + 2\mathcal{O}_{\rm VI}$ (LT Eq. (3.13), weights $\frac{P_X}{P_{\rm III(1)}}$);
  - $\mathcal{O}_C^z$ is $\mathcal{O}_C$ with $U(y') \to U(z)$ (the $y' \to z$ subtraction).

# 4 Three-$\rho$ rows: two-$\rho$ parts in singlet form

Each block is $\frac{1}{(2\pi)^3}\int \mathcal{K}|_{\rm merge}\,\mathcal{P}\,[\dots]$, to be taken as 2Re.

**Group I Row II.**

$$\begin{aligned}\frac{d^3N^{(2)}_{\rm II}}{d^3k} = \frac{1}{(2\pi)^3}\int \Big\{ &\mathcal{K}_{\rm II}\big|_{x\to x'}\mathcal{P}(x',y)\Big[-i\,N_c\,{\rm Tr}\big\{\big(U(y)-U(w)\big)U^\dagger(x')\big\} - i\,N_c\,{\rm Tr}\big\{U(w)U^\dagger(w')\big\} \\ &+ i\,{\rm Tr}\big\{T^eU(w')U^\dagger(x')T^eU(y)U^\dagger(w')\big\}\Big] \\ &+ \mathcal{K}_{\rm II}\big|_{y\to x'}\mathcal{P}(x',x)\Big[-i\,N_c\,{\rm Tr}\big\{\big(U(x')-U(w')\big)U^\dagger(w)\big\} + i\,N_c\,{\rm Tr}\big\{U(x)U^\dagger(x')\big\} \\ &- i\,{\rm Tr}\big\{T^eU(w')U^\dagger(x)T^eU(x')U^\dagger(w')\big\}\Big]\Big\}.\end{aligned} \tag{4.1}$$

**Group I Row III.**

$$\begin{aligned}\frac{d^3N^{(2)}_{\rm III}}{d^3k} = \frac{1}{(2\pi)^3}\int \Big\{ &\mathcal{K}_{\rm III}\big|_{y\to x}\mathcal{P}(x,x')\Big[\tfrac{i}{2}N_c\,{\rm Tr}\big\{\big(U(x')-U(w')\big)U^\dagger(x)\big\} + \tfrac{i}{2}\,{\rm Tr}\big\{T^eU(w')U^\dagger(x)T^eU(z)U^\dagger(w')\big\} \\ &- \tfrac{i}{2}\,{\rm Tr}\big\{T^eU(x)U^\dagger(x')T^eU(x')U^\dagger(z)\big\}\Big] \\ &+ \mathcal{K}_{\rm III}\big|_{x\to x'}\mathcal{P}(x',y)\Big[\tfrac{i}{2}N_c\,{\rm Tr}\big\{\big(U(y)-U(z)\big)U^\dagger(x')\big\} - \tfrac{i}{2}\,{\rm Tr}\big\{\big(U(y)-U(z)\big)U^\dagger(w')T^eU(w')U^\dagger(x')T^e\big\}\Big] \\ &+ \mathcal{K}_{\rm III}\big|_{y\to x'}\mathcal{P}(x',x)\Big[-\tfrac{i}{2}N_c\,{\rm Tr}\big\{U(x)U^\dagger(x')\big\} + \tfrac{i}{2}\,{\rm Tr}\big\{\big(U(x')-U(z)\big)U^\dagger(w')T^eU(w')U^\dagger(x)T^e\big\} \\ &+ \tfrac{i}{2}\,{\rm Tr}\big\{T^eU(x)U^\dagger(x')T^eU(x')U^\dagger(z)\big\}\Big]\Big\}.\end{aligned} \tag{4.2}$$

**Group I Row V, part 1.**

$$\begin{aligned}\frac{d^3N^{(2)}_{\rm V1}}{d^3k} = \frac{1}{(2\pi)^3}\int \Big\{ &\mathcal{K}_{\rm V1}\big|_{y\to x'}\mathcal{P}(x',y')\Big[i\,N_c\,{\rm Tr}\big\{U(x')U^\dagger(y')\big\} - i\,{\rm Tr}\big\{T^eU(w')U^\dagger(x')T^eU(y')U^\dagger(w')\big\} \\ &+ i\,{\rm Tr}\big\{T^eU(w')U^\dagger(x)T^eU(z)U^\dagger(w')\big\} - i\,{\rm Tr}\big\{T^eU(x)U^\dagger(x')T^eU(x')U^\dagger(z)\big\}\Big] \\ &+ \mathcal{K}_{\rm V1}\big|_{y'\to x'}\mathcal{P}(x',y)\Big[-i\,N_c\,{\rm Tr}\big\{U(x')U^\dagger(y)\big\} + i\,{\rm Tr}\big\{T^eU(w')U^\dagger(x')T^eU(y)U^\dagger(w')\big\}\end{aligned}$$

$$- i\,\mathrm{Tr}\{T^e U(w')U^\dagger(x)T^e U(z)U^\dagger(w')\} + i\,\mathrm{Tr}\{T^e U(x)U^\dagger(x')T^e U(x')U^\dagger(z)\}\Big]\Big\}. \tag{4.3}$$

With LT Eq. (3.4) the $\xi$ integral of $\mathcal{K}_{\rm V1}$ is in closed form (Ein), and $y \to x'$ or $y' \to x'$ only changes the transverse factor.

**$\mathcal{K}_{\rm IV}$ family** (Group I Row IV, Group III Rows I, II, Group II Row I part 1).

$$\begin{aligned}\frac{d^3N^{(2)}_{\rm IV}}{d^3k} = \frac{1}{(2\pi)^3}\int \Big\{ &\mathcal{K}_{\rm IV}\big|_{y\to x}\mathcal{P}(x,x')\Big[\tfrac{i}{2}N_c\,\mathrm{Tr}\{(U(x')-U(w'))U^\dagger(w)\} + \tfrac{i}{2}\,\mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(z)U^\dagger(w')\}\\ &- \tfrac{i}{2}\,\mathrm{Tr}\{T^eU(x)U^\dagger(x')T^eU(x')U^\dagger(z)\}\Big]\\ &+ \mathcal{K}_{\rm IV}\big|_{x\to x'}\mathcal{P}(x',y)\Big[\tfrac{i}{2}N_c\,\mathrm{Tr}\{(U(z)-U(w))U^\dagger(x')\} + \tfrac{i}{2}N_c\,\mathrm{Tr}\{U(w)U^\dagger(w')\} - \tfrac{i}{2}\,\mathrm{Tr}\{T^eU(w')U^\dagger(x')T^eU(z)U^\dagger(w')\}\Big]\\ &+ \mathcal{K}_{\rm IV}\big|_{y\to x'}\mathcal{P}(x',x)\Big[-\tfrac{i}{2}N_c\,\mathrm{Tr}\{U(w)U^\dagger(x')\} - \tfrac{i}{2}N_c\,\mathrm{Tr}\{U(x)U^\dagger(x')\} + \tfrac{i}{2}\,\mathrm{Tr}\{T^eU(x)U^\dagger(x')T^eU(x')U^\dagger(z)\}\\ &+ \tfrac{i}{2}\,\mathrm{Tr}\{T^eU^\dagger(w)U(x')T^eU^\dagger(x')U(z)\}\Big]\Big\}.\end{aligned} \tag{4.4}$$

**$\mathcal{K}_C$ family** (Group III Rows III part 1, V, VI). Line 1 and the first subtraction term ($\mathcal{O}_C$):

$$\begin{aligned}\frac{d^3N^{(2)}_{C}}{d^3k} = \frac{1}{(2\pi)^3}\int \Big\{ &(\mathcal{K}_C-\mathcal{K}_{C2})\big|_{y\to x}\mathcal{P}(x,x')\Big[-i\,\mathrm{Tr}\{(U(z)-U(w))U^\dagger(x')T^eU(x')U^\dagger(x)T^e\}\\ &+ i\,\mathrm{Tr}\{(U(z)-U(w))U^\dagger(y')T^eU(y')U^\dagger(x)T^e\}\Big]\\ &+ (\mathcal{K}_C-\mathcal{K}_{C2})\big|_{x\to x'}\mathcal{P}(x',y)\Big[-i\,N_c\,\mathrm{Tr}\{(U(y)-U(w))U^\dagger(x')\}\\ &+ i\,\mathrm{Tr}\{(U(x')-U(z))U^\dagger(y')T^eU(y')(U^\dagger(y)-U^\dagger(w))T^e\} + i\,\mathrm{Tr}\{(U(y)-U(w))U^\dagger(x')T^eU(x')U^\dagger(z)T^e\}\Big]\\ &+ (\mathcal{K}_C-\mathcal{K}_{C2})\big|_{y\to x'}\mathcal{P}(x',x)\Big[i\,N_c\,\mathrm{Tr}\{(U(x)-U(z))U^\dagger(x')\} - i\,\mathrm{Tr}\{(U(x)-U(z))U^\dagger(x')T^eU(x')U^\dagger(w)T^e\}\\ &- i\,\mathrm{Tr}\{(U(x)-U(z))U^\dagger(y')T^eU(y')(U^\dagger(x')-U^\dagger(w))T^e\}\Big]\Big\}.\end{aligned} \tag{4.5}$$

The second subtraction term ($+\mathcal{K}_{C2z}\mathcal{O}^z_C$, the $U(y') \to U(z)$ operator):

$$\begin{aligned}\frac{d^3N^{(2)}_{C,z}}{d^3k} = \frac{1}{(2\pi)^3}\int \Big\{ &\mathcal{K}_{C2z}\big|_{y\to x}\mathcal{P}(x,x')\Big[i\,N_c\,\mathrm{Tr}\{U(x)U^\dagger(z)\} - i\,\mathrm{Tr}\{(U(z)-U(w))U^\dagger(x')T^eU(x')U^\dagger(x)T^e\}\\ &- i\,\mathrm{Tr}\{T^eU(w)U^\dagger(z)T^eU(z)U^\dagger(x)\}\Big]\\ &+ \mathcal{K}_{C2z}\big|_{x\to x'}\mathcal{P}(x',y)\Big[-i\,N_c\,\mathrm{Tr}\{(U(y)-U(w))U^\dagger(x')\} - i\,N_c\,\mathrm{Tr}\{(U(y)-U(w))U^\dagger(z)\}\\ &+ i\,\mathrm{Tr}\{(U(y)-U(w))U^\dagger(x')T^eU(x')U^\dagger(z)T^e\} + i\,\mathrm{Tr}\{(U(y)-U(w))U^\dagger(z)T^eU(z)U^\dagger(x')T^e\}\Big]\\ &+ \mathcal{K}_{C2z}\big|_{y\to x'}\mathcal{P}(x',x)\Big[-i\,N_c\,\mathrm{Tr}\{U(w)U^\dagger(z)\} + i\,N_c\,\mathrm{Tr}\{U(x)U^\dagger(x')\}\\ &- i\,\mathrm{Tr}\{(U(x')-U(w))U^\dagger(z)T^eU(z)U^\dagger(x)T^e\} - i\,\mathrm{Tr}\{(U(x)-U(z))U^\dagger(x')T^eU(x')U^\dagger(w)T^e\}\Big]\Big\}.\end{aligned} \tag{4.6}$$

# 5 Two-$\rho$ rows in singlet form

Charges $(x', x)$ or $(x', y)$ as indicated; the kernels are those of LT. "LO-type" is the colour of LO, the CMW term, DGLAP, the running coupling and $\mathrm{LO}^{\rm lp}_\xi$.

$$\text{LO-type}: \quad \mathcal{P}(x',x)\,\mathrm{Tr}\{(U(x)-U(y))(U^\dagger(x')-U^\dagger(y'))\}, \tag{5.1}$$

$$H_4: \quad \mathcal{P}(x',x)\,\mathrm{Tr}\{(U(x)-U(z))(U^\dagger(x')-U^\dagger(z))\}, \tag{5.2}$$

$$\begin{aligned}\text{G1 V } 2a+2b \text{ (LT (3.6), first two lines)}: \quad &\mathcal{P}(x',y)\Big[N_c\,\mathrm{Tr}\{(U(x')-U(w'))U^\dagger(y)\} + \mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(z)U^\dagger(w')\}\\ &- \mathrm{Tr}\{T^eU(x)U^\dagger(x')T^eU(x')U^\dagger(z)\}\Big],\end{aligned} \tag{5.3}$$

$$\text{G1 V, the } \tfrac{11}{6} \text{ line}: \quad \mathcal{P}(x',y)\,N_c\,\mathrm{Tr}\{(U(x')-U(w'))(U^\dagger(y)-U^\dagger(z))\}, \tag{5.4}$$

$$\begin{aligned}\text{G2 I part 2 } (2a\text{–}2f, \text{ all}): \quad &\mathcal{P}(x',x)\Big[N_c\,\mathrm{Tr}\{U(w)U^\dagger(w')\} + N_c\,\mathrm{Tr}\{U(x)U^\dagger(x')\} - \mathrm{Tr}\{T^eU(w)U^\dagger(x')T^eU(z)U^\dagger(w)\}\\ &- \mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(w')\}\Big],\end{aligned} \tag{5.5}$$

$$\begin{aligned}\text{G3 III(2), line 1, } (x'-y') \text{ of line 2}: \quad &\mathcal{P}(x',x)\Big[-N_c\,\mathrm{Tr}\{(U(x')-U(y'))U^\dagger(x)\} + \mathrm{Tr}\{T^eU(w)U^\dagger(x')T^eU(x')U^\dagger(z)\}\\ &- \mathrm{Tr}\{T^eU(w)U^\dagger(y')T^eU(y')U^\dagger(z)\}\Big],\end{aligned} \tag{5.6}$$

$$\begin{aligned}\text{G3 III(2), } (x'-z) \text{ of line 2, line 3}: \quad &\mathcal{P}(x',x)\Big[N_c\,\mathrm{Tr}\{(U(x)-U(w))U^\dagger(z)\} - N_c\,\mathrm{Tr}\{U(x)U^\dagger(x')\}\\ &+ \mathrm{Tr}\{T^eU(w)U^\dagger(x')T^eU(x')U^\dagger(z)\}\Big].\end{aligned} \tag{5.7}$$

The colour of Group II Row I part 2 is the same ($\mathcal{O}_{58}$) for all of $2a$–$2f$, so that row is $\frac{1}{(2\pi)^3}\int\big(\sum_{2a}^{2f}\text{kernels}\big)\,\mathcal{P}(x',x)[\dots]$.

# 6 Final result in colour-singlet form (Gaussian projectile)

With LO in the singlet form (5.1):

$$\begin{aligned}\frac{d^3N_g}{d^3k} &= \mathrm{LO}\big|_{\alpha_s\to\alpha_s(c_0/|y'-y|)} + \frac{\alpha_s N_c}{2\pi}\Big(\frac{67}{9}-\frac{\pi^2}{3}\Big)\mathrm{LO} + \log\frac{k^+}{\Lambda}\,\big(\mathrm{JIMWLK}\otimes\mathrm{LO}\big) + \log\frac{\vee}{k^+}\,\big(\mathrm{BFKL}\otimes\mathrm{LO}\big) + \mathrm{DGLAP\ (LT\ (3.8))} \\ &\quad - H_4 + \frac{\alpha_s}{2\pi}2N_c\int_0^1 d\xi\,\frac{1-\xi}{\xi}\mathrm{LO}_\xi^{\mathrm{lp}} + \frac{d^3N_{\mathrm{G2\,I(2)}}}{d^3k} \\ &\quad + 2\,\mathrm{Re}\Big[\frac{d^3N^{(2)}_{\mathrm{II}}}{d^3k} + \frac{d^3N^{(2)}_{\mathrm{III}}}{d^3k} + \frac{d^3N^{(2)}_{\mathrm{V1}}}{d^3k} + \frac{d^3N_{\mathrm{V}\,2a+2b}}{d^3k} + \frac{d^3N^{(2)}_{\mathrm{IV}}}{d^3k} + \frac{d^3N^{(2)}_{C}}{d^3k} + \frac{d^3N^{(2)}_{C,z}}{d^3k} + \frac{d^3N_{\mathrm{G3\,III(2)}}}{d^3k}\Big].\end{aligned} \tag{6.1}$$

Every term is $\int \mathcal{K}\,\mathcal{P}(u,v)\times$(traces). The three-$\rho$ rows enter only through their merged (two-$\rho$) parts.

# 7 Checks (`singlet3.json`)

| check | pieces | max deviation |
|---|---|---|
| trace form = direct contraction $\rho^a\rho^b\to\delta^{ab}$ (3 random general SU(3) $U$'s) | all two-$\rho$ rows; all merges of II, III, V1, IV, $C$, $C_z$ | $7.8\times10^{-16}$ |
| after combining into differences $U(p)-U(q)$ | same | $5.2\times10^{-16}$ |
| pair rule = reordering engine (`../two_rho/core.symmetrize`) | II, III, V1, IV, $C$, $C_z$ | $2.4\times10^{-16}$ |
| $\mathcal{O}_C^z=\mathcal{O}_C$ at $U(y')=U(z)$ | Group III Rows III(1), V, VI | 0 |
| colour operators = your file (for $U=U^T$) and first principles (general $U$) | all three-$\rho$ rows | `gsrc.json` (earlier) |